

Research Article

Soil Characteristics, Growth, Yield, Mineral Composition and Proximate Content of Soybeans Under Various Tillage Techniques

Aruna Olasekan Adekiya ¹, Olajire Damilola Adedokun,¹ and Ojo Timothy Adebisi²

¹Agriculture Programme, College of Agriculture, Engineering and Science, Bowen University, Iwo, Nigeria

²College of Agricultural Sciences, Landmark University, PMB 1001, Omu-Aran, Kwara State, Nigeria

Correspondence should be addressed to Aruna Olasekan Adekiya; adekiya2009@yahoo.com

Received 13 May 2024; Accepted 23 September 2024

Academic Editor: Vera Popovic

Copyright © 2024 Aruna Olasekan Adekiya et al. This is an open access article distributed under the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

Different tillage methods can have varying effects on soil properties and crop quality. Therefore, field experiments were conducted at the Teaching and Research Farm of Landmark University, located in Omu-Aran, Kwara State. The objectives were to investigate the impacts of various tillage methods on soil characteristics and the growth, yield, mineral and proximate content of soybean (*Glycine max* (L.) Merr.). Based on these objectives, it was hypothesized that soil properties, growth, yield, mineral and proximate contents of soybean would react differently to different tillage methods. Consequently, experiments were conducted to validate this hypothesis. The treatments were four tillage methods: manual clearing (MC), ploughing (P), ploughing plus harrowing (P + H) and manual ridging (MR), replicated three times in a randomized complete block design. Data on the initial soil properties (bulk density, particle size, pH, organic matter, N, P, K, Ca and Mg), along with the chemical properties of the soil on a per-plot basis at the end of the experiment, were collected. Additionally, data on soil bulk density, porosity, moisture content and temperature were collected during the vegetative growth of the soybean. Soybean plant height and the number of branches were recorded at the midflowering stage, while grain yield and the number of pods were recorded at harvest. Soybean seeds were later analysed for their mineral and proximate contents using standard methods. P + H resulted in lower soil bulk density than other tillage methods and led to higher growth, yield, proximate and mineral contents of soybean. In MC, bulk density, soil moisture content, soil organic matter, N, P, K, Ca and Mg were significantly higher and temperature lower than in other tillage methods; yet, these advantages did not translate into increased yield and quality due to the higher bulk density. The growth, yield and quality of soybean were significantly influenced by soil bulk density rather than by soil chemical properties. Using the mean of the two sites, relative to MC, P and MR, P + H increased the grain yield of soybean by 41.3%, 24.4% and 13.7%, respectively. These findings underscore the importance of appropriate tillage practices for successful soybean cultivation in similar agroecological zones. For small-scale operations, MR is suggested, while P + H is recommended for large-scale soybean production in the studied area. Future studies could include additional variables, such as economic analysis, long-term soil health impacts or the effects of tillage methods under varying climate conditions.

Keywords: bulk density; mineral content of soybean; proximate content of soybean; soil chemical properties; soybean; tillage

1. Introduction

Soil management plays a pivotal role in enhancing agricultural practices and boosting crop production [1]. Among the methods employed in soil management, tillage stands out as a significant approach to maintain soil fertility. The choice of tillage methods is crucial in agricultural

endeavours as it directly impacts soil attributes and subsequently influences crop development and yield. Particularly in regions characterized by humid tropical climates, where many farmers face economic constraints and fertilizers are costly, implementing soil working and tillage techniques can serve as a temporary substitute for fertilizer application [2].

The choice of tillage method significantly can alter the physical, chemical and biological attributes of the soil matrix [3]. Traditional tillage involves extensive soil disturbance through ploughing and harrowing (P + H), altering soil structure by affecting physical properties like bulk density, penetration resistance and moisture content. This disturbance and fragmentation lead to a finer, looser soil structure [4]. In contrast, conservation tillage methods such as no-till (manual clearing [MC]) and reduced tillage aim to minimize soil disturbance, preserving soil structure and organic matter (OM). These practices often result in reduced pore space [5], increased soil strength [6] and the formation of stable aggregates. Conservationally tilled soil typically features a more continuous pore network due to activities like earthworms, root channels and vertical cracks [7]. Thus, conservation tillage tends to mitigate the disruption of continuous pores. Each tillage approach engenders distinct soil conditions, affecting moisture retention, nutrient availability, soil compaction and microbial activity, thereby shaping the soil environment in which crops thrive.

Previous studies have demonstrated that tillage practices play a crucial role in shaping soil properties and influencing crop growth and yield. Various tillage methods—such as zero tillage (ZT), manual ridging (MR), manual mounding (MM), single P followed by harrowing and double P + H—were assessed for their impact on soil's physical and chemical characteristics, as well as the fresh rhizome yield of ginger (*Zingiber officinale* Roscoe) [8]. Among the methods, both single and double P followed by harrowing were associated with increased soil bulk density, which corresponded with reduced levels of soil OM, nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg) and lower ginger rhizome yields. Conversely, ZT was found to enhance soil bulk density, moisture content, temperature and the levels of OM, N, P, K, Ca and Mg more effectively than other tillage methods, leading to greater plant height and higher ginger rhizome yields.

Between 2008 and 2010, a study was conducted at two locations—Owo and Akure in southwest Nigeria—to evaluate the impact of five tillage methods on soil properties, leaf nutrient content, growth and tuber yield of yam (*Dioscorea rotundata* Poir.) on Alfisols. The tillage methods tested were ZT, MR, MM, P + H and ploughing plus harrowing followed by ridging (P + H + R) [9]. When comparing the effects on yam tuber yield, P + H + R showed significant increases over MR, MM, P + H and ZT. Specifically, P + H + R improved tuber yield by 12.3%, 12.8%, 34.9% and 50.7% at Site A and by 12.9%, 13.5%, 25.2% and 44.5% at Site B. This method was the most beneficial and is recommended for yam cultivation. The results indicated that P + H + R produced the highest mean fruit weight (8.16 t/ha), along with superior plant height and tap root length. This was followed by MM (7.05 t/ha), P + H (6.24 t/ha) and P (5.81 t/ha), with MC yielding the lowest (5.03 t/ha).

Similarly, a study conducted in Kumasi, Ghana, evaluated the impact of four tillage practices on the performance of Asontem cowpea (*Vigna unguiculata* (L) Walp) [10]. The treatments included disc P only, disc P followed

by disc harrowing, disc harrowing only and no tillage. The combination of disc P followed by disc harrowing led to the best results, showing the highest seedling emergence rate, tallest plants, thickest stem girth and the greatest number of leaves.

These findings underscore that the choice of tillage method can have a significant impact on soil characteristics and plant development. The effects include changes in soil structure, moisture retention, OM content, erosion control, weed and pest management, microbial activity and root development. The optimal tillage approach often varies depending on the specific soil type, climate and crop needs [11].

Soybean (*Glycine max* (L.) Merr.) holds significant importance as a staple legume in the diets of Nigerians. Originating from Eastern Asia, where it has been a dietary mainstay for centuries [12], soybean is renowned for its rich protein and oil content, alongside high levels of essential amino acids like lysine, leucine and lecithin. With approximately 40%–45% protein and 18%–22% oil, soybean also serves as a valuable source of vitamins and minerals [13]. It is hailed as the prospective protein source of the future due to its provision of high-quality protein and oil [12]. The oil extracted from soybean finds extensive use in food applications, including margarine, cooking oils and salad dressings [14].

The literature presents substantial evidence, indicating that tillage enhances the growth and yield of soybeans [15, 16]. However, given that soybeans, like other crops, are anticipated to respond differently to various tillage methods concerning growth, yield and quality, this aspect warrants investigation, particularly in Nigeria where data on the effects of different tillage methods on the mineral and proximate contents of soybeans are lacking. Previous studies have primarily focused on the impact of tillage methods on soybean growth and yield, neglecting to consider seed quality. For instance, Lasisi and Aluko [12] reported that conventional tillage (P and P + H) yielded significantly better soybean growth responses (such as plant height, leaf area, number of pods per stand and seed yield) than conservation tillage methods in Nigeria. Similarly, in Ethiopia, Adugna [4] conducted an experiment aimed at assessing the effect of different tillage methods on the crop yield of both sole maize and soybeans, as well as their intercropping. The study found that soybean yield was higher under conventional tillage than no tillage. Additionally, a study on the impact of various tillage practices on soybean productivity and soil moisture dynamics during the 2019 kharif season [13] revealed favourable effects of tillage practices on soybean growth, yield parameters and soil moisture content under rainfed conditions.

Furthermore, the influence of tillage practices extends beyond soil properties to encompass soybean growth dynamics and nutritional composition. Variations in soil structure, nutrient distribution and microbial activity directly influence root development, nutrient uptake and, ultimately, soybean biomass accumulation and yield. Additionally, alterations in soil composition may impact the mineral and proximate content of soybeans,

potentially modulating their nutritional quality and functional attributes. Therefore, to have a full approach on the response of soybean performance to different tillage methods, data on the quality of such soybean seed are imperative. Hence, the objectives of this study were to determine the effect of different tillage methods on soil properties and soya bean growth, yield, mineral and proximate contents. Based on these objectives, it was hypothesized that soil properties, growth, yield, mineral and proximate contents of soybean will react differently to different tillage methods. Therefore, experiments were conducted to validate this hypothesis.

2. Materials and Methods

2.1. Site Description and Tillage Treatments. During the 2020 cropping season, field experiments were conducted at the Teaching and Research Farm of Landmark University, located in Omu-Aran, Kwara State. The aim was to investigate the impacts of various tillage methods on soil characteristics and the growth, yield, mineral and proximate content of soybeans. Landmark University is situated between latitudes $8^{\circ} 7' 26.21388''$ and $5^{\circ} 5' 0.1788''$, with an annual rainfall of approximately 1300 mm and an average annual temperature of 32°C . The soil type in the experimental area is classified as either Oxic Haplustalf or Luvisol, both falling under the category of Alfisols [17]. Prior to the experiment, the land had been fallow for two years following previous cultivation of maize and cassava. The predominant weeds observed at the study sites included Siam weed (*Chromolaena odorata* L. King and Robinson), Mexican sunflower (*Tithonia diversifolia*) and water leaf (*Talinum triangulare* Jacq. Willd.), alongside interspersed shrubs.

At each site, positioned approximately 50 meters apart, four distinct tillage methods were implemented for the experiments. These treatments were replicated three times in a randomized complete block design. They included the following:

- (1) MC: This involved MC using a cutlass, with weeds removed from the plots before planting on a flat surface without any primary or secondary tillage operation.
- (2) P: Soil was ploughed to a depth of 0.20 meters once, utilizing a tractor-mounted disc plough.
- (3) P + H: Soil underwent P and harrowing to a depth of 0.20 meters once, employing both a tractor-mounted disc plough and harrow.
- (4) MR: The ridge was formed by heaping the surface layer of soil using a traditional hoe, following the removal of cleared weeds from the plots.

To minimize interference between blocks, they were positioned 4 meters apart, while plots were spaced 3 meters apart. Prior to tillage operations for treatments P and P + H, an initial clearing of the plots was conducted. Each plot measured 12 meters by 10 meters. The tillage treatments were executed in July 2020.

2.2. Crop Establishment. On 25 July 2020, the sowing of soybean seeds of the TGX 1448-2E variety took place. Three seeds were planted with a spacing of 5 centimetres within rows and 60 centimetres between rows. Thinning was performed two weeks after sowing to retain only one plant per stand, resulting in a population of 4000 plants per plot. Throughout the experiment, three manual weeding sessions were conducted at intervals of 4, 8 and 12 weeks after sowing.

2.3. Soil Sampling and Analysis. Before initiating the experiment in 2020, undisturbed soil samples were collected from ten designated points at each of the two sites and analysed for bulk density, particle size and chemical properties. Also, disturbed soil samples were collected from each plot upon soybean harvest, ranging from the surface to a depth of 15 cm, and underwent chemical analysis as shown in Table 1.

2.4. Determination of Soil Physical Properties. Four weeks after tillage (about 3 1/2 weeks after soybean sowing), during the vegetative growth of soybean crops, the assessment of specific soil physical properties across all plots at the research sites commenced. This was done at two weeks of intervals on four occasions, for each site. Utilizing a steel coring tube, five undisturbed samples were obtained from each plot, ranging from 0 to 15 cm in depth. These samples were utilized to evaluate bulk density and gravimetric moisture content, following the methodology outlined previously [18]. Soil temperature was also monitored bi-weekly over four occasions at each site, with measurements taken at 15:00 h using a soil thermometer inserted to a depth of 15 cm. Five readings were recorded per plot during each sampling event, and the mean temperature was calculated.

2.5. Determination of Growth and Yield Parameters

2.5.1. Growth Parameters. Growth parameters were assessed during the midflowering stage of the soybean plants, which occurred approximately 60 days after sowing. The study measured various plant parameters, including plant height and the number of branches per plant. For this analysis, ten plants were randomly chosen from each plot.

Plant height was measured from the ground level to the tip of the terminal bud using a meter rule. The heights of ten plants per plot were recorded, and their average values were calculated to provide a representative measure for each plot.

2.5.2. Yield Parameters. Upon reaching maturity (110 days after sowing), the soybean pods were harvested, and the number of pods harvested was counted and recorded for each plot and treatment. Subsequently, threshing was performed to separate the seeds from the pods. The threshed soybean seeds were weighed using a precise weighing balance, and the values were recorded on a per-plot basis.

TABLE 1: Soil analysis before and after imposition of tillage treatments.

Soil characteristics	Method
Bulk density	By employing steel coring tubes measuring 4 cm in diameter and 15 cm in height. These samples were then subjected to an oven set at 100°C for a duration of 24 hours to determine the bulk density [18]
Particle size	Particle size analysis was conducted through the hydrometer method [19], and the textural class was determined utilizing a textural triangle [20, 21]
Organic carbon	Soil organic carbon content was determined following the Walkley and Black procedure utilizing the dichromate wet oxidation method [22]. Organic matter content was calculated by multiplying the carbon content by 1.724
Total N	Total nitrogen content was determined utilizing the micro-Kjeldahl digestion method [23]
Available P	Available phosphorus was assessed through Bray-1 extraction followed by molybdenum blue colorimetry [24]
Exchangeable K, Ca and Mg	Exchangeable potassium, calcium and magnesium were extracted using 1 M ammonium acetate [25]. Potassium levels were determined using a Jenway PFP7 flame photometer, while calcium and magnesium were determined employing the EDTA titration method [26]
Soil pH	Soil pH was measured using a glass pH meter at a soil-to-water ratio of 1 : 2 [27]

2.6. Determination of Proximate Composition. The proximate compositions of ground soybean seeds were determined through established methods [28]. Moisture content was determined by drying 2 grams of ground soybean seeds in a crucible at 105°C until a constant weight was achieved, with the percentage loss representing moisture content [28]. Ash content was determined by subjecting 2 grams of dried milled soybean seeds to a muffle furnace at 550°C for 6 hours. The resulting ash residue in the crucible was cooled, weighed and expressed as ash content. Protein content was assessed using the Kjeldahl method, involving sample digestion and nitrogen content quantification to calculate protein levels [28]. Crude fat was determined by the method of Udo and Oguwale [29]. Crude fat was determined by the method of Udo and Oguwale [29]. Crude fibre estimation involved acid and alkaline digestion methods utilizing 2.00 grams of each sample with 20% H₂SO₄ and NaOH solution [30]. Carbohydrate content was determined through estimation using the arithmetic difference method [31]:

$$\begin{aligned} \% \text{carbohydrate} = 100 - (\% \text{moisture} + \% \text{fat} \\ + \% \text{ash} + \% \text{fibre} + \% \text{protein}). \end{aligned} \quad (1)$$

2.7. Determination of Mineral Content. Mineral analysis involved ashing of 10 grams of ground soybean seeds, for 5 hours at 550°C in a cleaned porcelain crucible. The resulting ash residue was dissolved in a solution of 5 millilitres of HNO₃/HCl (in a 1 : 2 ratio) and gently heated on a hot plate until brown fumes dissipated, yielding a white solution. Each crucible's solution was then filtered into a 100-millilitre volumetric flask and diluted to 100 millilitres with deionized water. Cations (including Na, K, Mg, Ca, P, Zn and Fe) were quantified using a flame atomic absorption spectrophotometer [32].

2.8. Statistical Analysis. Data collected from each experiment underwent analysis of variance (ANOVA) utilizing the Genstat statistical package [33]. This was carried out to ascertain the impact of treatments on various aspects including soil physical and chemical properties, soybean growth and yield, as well as mineral and proximate contents.

3. Results

3.1. Initial Soil Fertility Status of the Experimental Site. Table 2 shows the results regarding the initial fertility status of the two sites before the implementation of tillage treatments. The surface texture at both experimental sites was identified as sandy loam. These sites exhibited relatively high bulk density. However, they were characterized by poor fertility, featuring low levels of organic carbon (OC), nitrogen (N), phosphorus (P) and calcium (Ca), although potassium (K) and magnesium (Mg) levels were deemed adequate based on the critical values set for crop production in Nigeria [34], which include 3.0% OM, 0.20% N, 10.0 mg kg⁻¹ P, 0.16–0.20 cmol kg⁻¹ K, 2.0 cmol kg⁻¹ Ca and 0.40 cmol kg⁻¹ Mg.

3.2. Effect of Tillage Methods on Soil Physical Properties. The results of the effects of various tillage methods on soil physical properties are presented in Table 3. P + H significantly produced lowest soil bulk density and highest porosity compared with other treatments at both sites. Also, at both sites, MC has the highest moisture content and bulk density and recorded the lowest porosity and soil temperature. MR has the highest soil temperature and lowest moisture content, although these values were not significantly different from that produced by P + H. The increasing order of soil bulk density among the various tillage methods was as follows: P + H < MR < P < MC. The reverse order was the porosity. The increasing order of soil moisture content among the various tillage methods was as follows: MR = P + H < P < MC, while the increase order of soil

TABLE 2: Initial fertility status of the two sites before cropping.

Property	Site A	Site B
Sand (%)	68.1	68.6
Silt (%)	15.7	15.9
Clay (%)	16.2	15.2
Textural class	Sandy loam	Sandy loam
Bulk density (g/cm ³)	1.58	1.56
Porosity (%)	43.3	41.1
Organic carbon (%)	1.63	1.66
Total N (%)	0.19	0.19
C:N ratio	8.58	8.74
Available P (mg/kg)	9.0	9.0
Exchangeable K (coml/kg)	0.17	0.16
Exchangeable Ca (coml/kg)	1.61	1.61
Exchangeable Mg (coml/kg)	0.83	0.85

TABLE 3: Effect of tillage methods on soil physical properties.

	Moisture content (%)		Bulk density (g/cm ³)		Porosity (%)		Temperature (°C)	
	Site A	Site B	Site A	Site B	Site A	Site B	Site A	Site B
MC	20.80a	19.60a	1.59a	1.57a	40.0d	40.7d	29.1d	29.6d
P	18.40b	18.10b	1.47b	1.49b	44.5c	43.8c	30.8cd	30.8cd
P + H	15.81c	15.96c	1.29d	1.29d	51.3a	51.3a	32.8ab	31.9ab
MR	16.10c	16.20c	1.38c	1.39c	47.9b	47.5b	34.2a	33.8a

Note: Values followed by similar letters under the same column are not significantly different at $p = 0.05$ according to Duncan's multiple range test.

Abbreviations: MC = manual clearing; MR = manual ridging; P = ploughing; P + H = ploughing + harrowing.

temperature was as follows: MC = P < P + H = MR. Using the mean of the two sites, relative to MC, P and MR, P + H reduced soil bulk density by 58%, 48% and 38%, respectively.

3.3. Effect of Tillage Methods on Soil Chemical Properties.

The results of the effect of tillage methods on soil chemical properties are presented in Table 4. Results showed that MC increased soil pH, OC, N, P, K, Ca and Mg contents significantly relative to other tilled treatments (although MC and P have values of pH, OC, N, K, Ca and Mg that are statistically similar). Tilled soils (P + H and MR) have lower soil chemical properties relative to untilled soil (MC). Conventionally tilled soil (P + H) seems to have the least values of soil chemical properties, and its values are not statistically different from that produced by MR except for P, Ca and Mg. The increasing order of soil chemical properties exhibited by the different tillage systems was as follows: P + H < MR < P < MC.

3.4. Effect of Tillage Methods on Growth and Yield of Soybean.

Table 5 shows the results of the effects of different tillage methods on the growth and yield of soybean. Tilled soils (P, P + H and MR) increased growth (plant height and number of branches) and yield (number of pod and grain yield) of soybean relative to untilled soil (MC). Among all treatments, MC has the least value. P + H consistently outperforms other methods in terms of plant height, number of branches per plant, number of pods per plant and grain yield across both Site A and Site B. The increasing order of growth and yield of soybean was as follows: MC < P < MR < P + H. Using the

mean of the two sites, relative to MC, P and MR, P + H increased the grain yield of soybean by 41.3, 24.4 and 13.7%, respectively.

3.5. Effect of Tillage Methods on Qualities (Proximate and Mineral Contents) of Soybean.

The effect of different tillage methods on the proximate and mineral contents of soybean are presented in Figures 1, 2, 3, 4, 5 and 6 and Table 6, respectively. Tilled treatments (P, P + H and MR) increased protein, crude fat, ash and fibre and reduced the carbohydrate contents of soybean relative to untilled treatment (MC). P + H increased the protein, crude fat, ash and fibre and reduced the carbohydrate contents of soybean relative to other tillage treatments. These values of P + H are not statistically different from that produced by MR. Also, all tilled treatments (P, P + H and MR) have statistically similar values for crude fat. MC increased moisture and carbohydrate contents of soybean significantly relative to other tillage methods. Also, tilled soils increased the mineral contents (Ca, Mg, K, Fe, Na, P and Zn) (Table 6) of soybean relative to untilled soil (MC) except in some cases where the values produced by P were not significantly different from that of MC. P + H has the highest value of mineral content, and the values were not however statistically different from that produced by MR.

4. Discussion

MC soil exhibited a significantly higher bulk density in contrast to other tillage techniques. This increased bulk density in MC plots may stem from the absence of tillage

TABLE 4: Effect of tillage methods on soil chemical properties.

	pH (water)		SOM (%)		N (%)		P (mg/kg)		K (cmol/kg)		Ca (cmol/kg)		Mg (cmol/kg)	
	Site A	Site B	Site A	Site B	Site A	Site B	Site A	Site B	Site A	Site B	Site A	Site B	Site A	Site B
MC	5.61a	5.59a	2.70a	2.71a	0.18a	0.18a	8.9a	8.9a	0.16a	0.17a	1.58a	1.60a	0.81a	0.82a
P	5.55ab	5.53ab	2.61ab	2.58ab	0.17ab	0.17ab	8.1b	8.0b	0.15ab	0.16ab	1.50a	1.51a	0.75ab	0.73ab
P + H	5.09c	5.10c	2.36d	2.31d	0.15c	0.16b	6.1d	6.3d	0.13c	0.13d	1.20c	1.24c	0.55d	0.57d
MR	5.45bc	5.40bc	2.44cd	2.50cd	0.15c	0.16b	6.8c	7.1c	0.14bc	0.14cd	1.36b	1.37b	0.60c	0.62c

Note: Values followed by similar letters under the same column are not significantly different at $p = 0.05$ according to Duncan’s multiple range test. Abbreviations: MC = manual clearing; MR = manual ridging; P = ploughing; P + H = ploughing + harrowing.

TABLE 5: Effect of tillage methods on growth and yield of soybean.

	Plant height (cm)		Number of branches/ plants		Number of pods/ plants		Grain yield (kg/ha)	
	Site A	Site B	Site A	Site B	Site A	Site B	Site A	Site B
MC	50.1d	51.6d	5.1c	5.2c	35.1d	34.6d	660.5d	648.1d
P	56.1c	57.2c	5.8b	6.0b	48.7c	48.9c	740.3c	746.6c
P + H	70.8a	71.5a	8.1a	5.4a	71.1a	71.6a	920.8a	928.5a
MR	62.1b	63.3b	6.1b	6.4b	62.6b	63.1b	810.9b	815.6b

Note: Values followed by similar letters under the same column are not significantly different at $p = 0.05$ according to Duncan’s multiple range test. Abbreviations: MC = manual clearing; MR = manual ridging; P = ploughing; P + H = ploughing + harrowing.

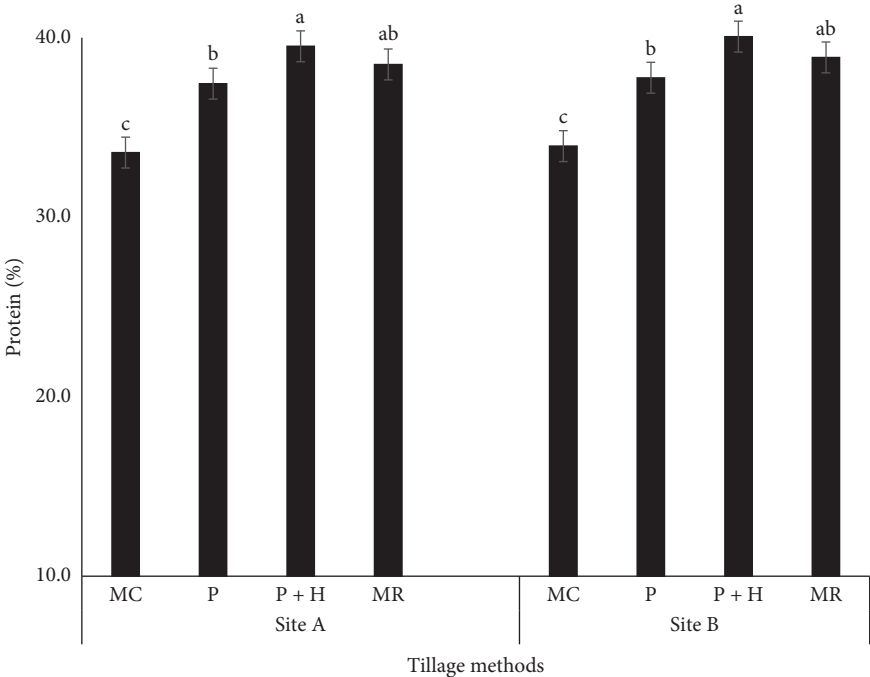


FIGURE 1: Effect of tillage methods on protein content of soybean. MC = manual clearing; P = ploughing; P + H = ploughing plus harrowing; MR = manual ridging. Vertical bars show standard error of paired comparisons; bars marked with different letters show means significantly different at 5% level using Duncan’s multiple range test.

practices and subsequent compaction. It suggests that leaving soil untilled and exposed to rainfall induces compaction over time. Prior studies by Bankole and Ojeniyi [35], as well as Agbede [11], also noted elevated bulk density in manually cleared areas compared to tilled soils in south-western Nigeria. Similarly, Ike [36] reported higher bulk density in untilled soils than tilled ones in Nigeria’s savanna zone. The comparatively lower bulk density observed in

P + H, P and MR could be attributed to the soil-loosening effects of tillage. Particularly, P + H displayed higher total porosity values than untilled soil, confirming the inverse relationship between bulk density and total porosity as documented by Dohahue, Miller and Shickluna [37].

MC soil demonstrated higher moisture content and lower temperature than tilled soils, likely due to the OM acting as mulch on the soil surface, reducing temperature

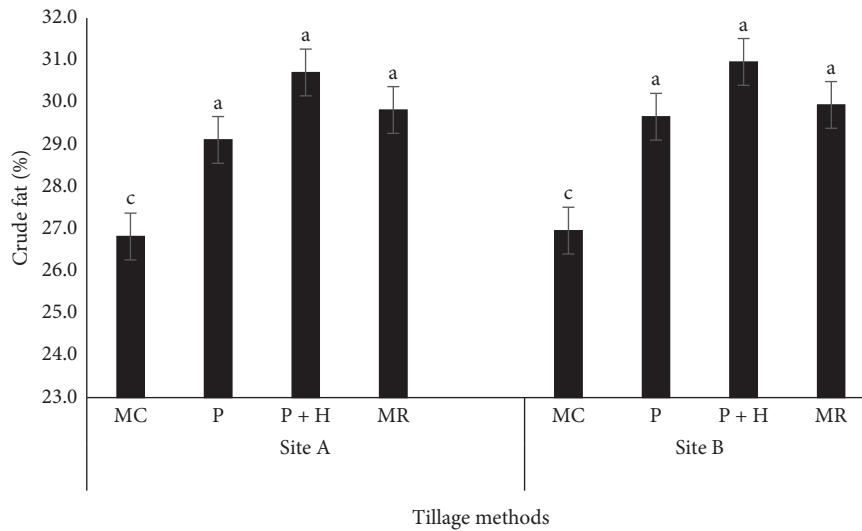


FIGURE 2: Effect of tillage methods on crude fat content of soybean. MC = manual clearing; P = ploughing; P + H = ploughing plus harrowing; MR = manual ridging. Vertical bars show standard error of paired comparisons; bars marked with different letters show means significantly different at 5% level using Duncan's multiple range test.

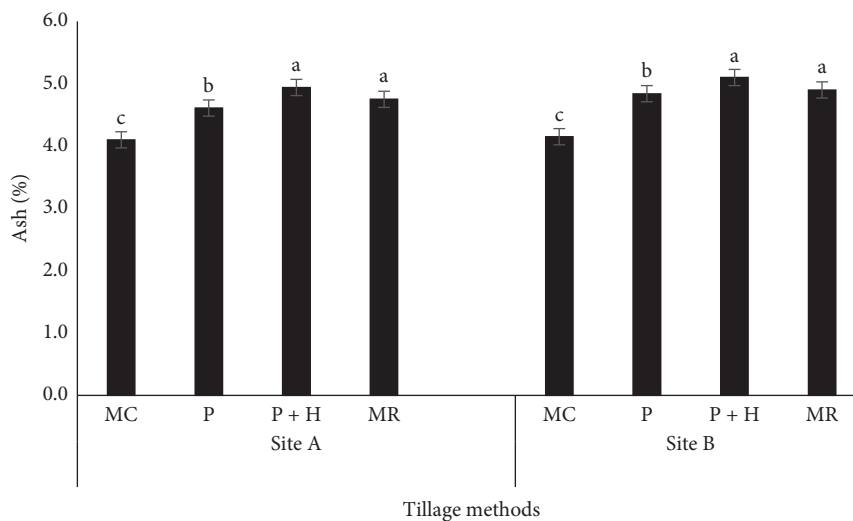


FIGURE 3: Effect of tillage methods on ash content of soybean. MC = manual clearing; P = ploughing; P + H = ploughing plus harrowing; MR = manual ridging. Vertical bars show standard error of paired comparisons; bars marked with different letters show means significantly different at 5% level using Duncan's multiple range test.

and water loss through evaporation as suggested by Ojeniyi, Abiwonninu and Odedina [38]. The higher moisture content in MC soil could also be linked to its higher density and consequent lower porosity, reducing the rate of evaporation. Ojeniyi [39] similarly observed that volumetric moisture content increased with bulk density in sandy soils of southwest Nigeria, thus explaining the relatively high moisture content in untilled manually cleared soils. Although not significantly different from P + H, MR produced the highest soil temperature. This could be attributed to ridging exposing soil more to radiation and increasing the evaporative loss of soil water, as noted by Lal [40] and Ojeniyi, Adetoro and Odedina [41].

The decline in soil pH values was less pronounced in MC plots compared to other tillage methods. This decline could be attributed to the loss of OM and cation exchange capacity. This aligns with the findings of Ndaeyo, Aiyelari and Agboola [42]. Additionally, conventional tillage plots may experience a decrease in pH due to soil erosion and leaching of base elements, particularly Ca and Mg, resulting in soil acidity. MC exhibited the highest levels of SOM, N, P, K, Ca and Mg across both sites compared to other treatments, likely attributable to the presence of OM. The decline in nutrient reserves observed in tilled soils, particularly in the P + H treatment, could be attributed to the frequent use of tractors (twice) during land preparation, which may have facilitated soil erosion, preferentially

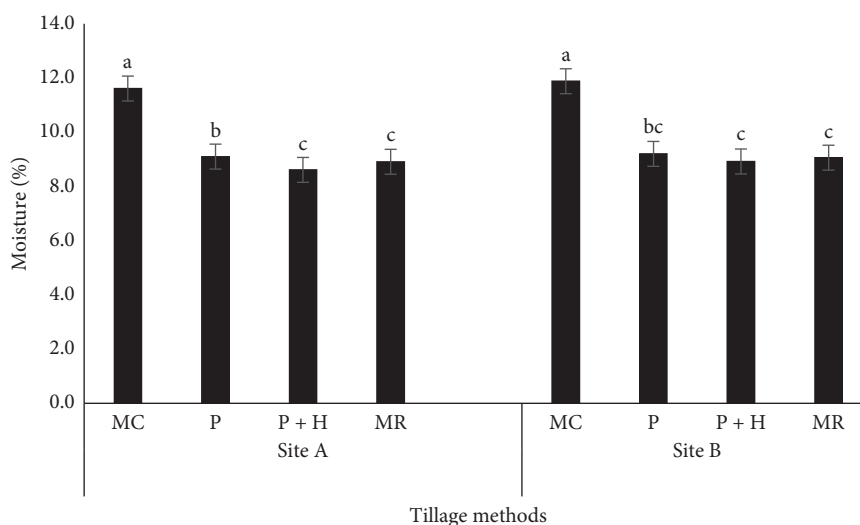


FIGURE 4: Effect of tillage methods on moisture content of soybean. MC=manual clearing; P=ploughing; P + H=ploughing plus harrowing; MR=manual ridging. Vertical bars show standard error of paired comparisons; bars marked with different letters show means significantly different at 5% level using Duncan's multiple range test.

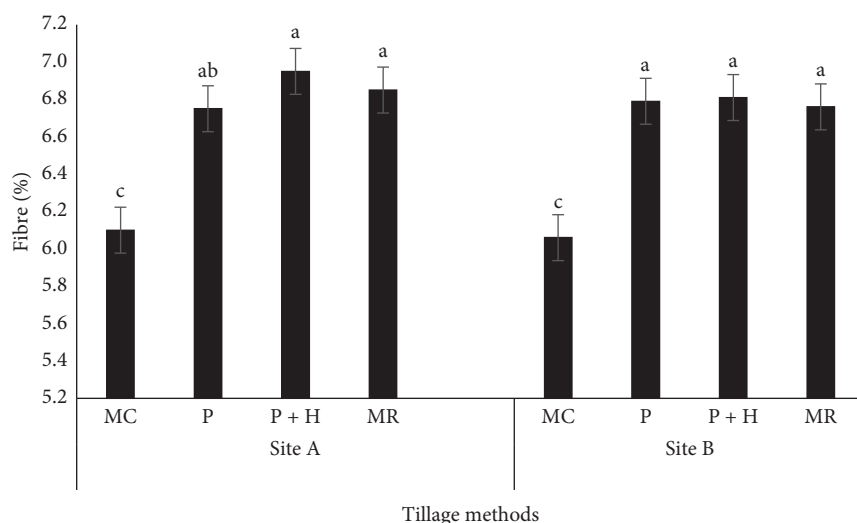


FIGURE 5: Effect of tillage methods on fibre content of soybean. MC=manual clearing; P=ploughing; P + H=ploughing plus harrowing; MR=manual ridging. Vertical bars show standard error of paired comparisons; bars marked with different letters show means significantly different at 5% level using Duncan's multiple range test.

removing colloidal fractions with high “enrichment ratio” [43, 44], leading to a gradual depletion of nutrient reserves. Tillage resulted in a reduction in soil OM and nutrient content compared to nontillage, such as MC, with P + H exhibiting the lowest values. The increased soil porosity associated with tillage should have promoted the oxidation and subsequent loss of OM. As OM serves as a natural reservoir of nutrients, its loss implies a decrease in soil fertility and nutrient levels. Additionally, tillage techniques, such as hoeing or P, invert the soil, disturbing the concentration of nutrients and OM in the topsoil, further diminishing its fertility. Moreover, the enhanced soil porosity from tillage facilitates the leaching of nutrients into the subsoil. The detrimental effect of tillage on soil fertility is exacerbated with increased tillage intensity [45].

The relatively low levels of SOM, N, P, K, Ca and Mg observed in the P + H soil in contrast to the MC soil might stem from the inversion of topsoil during land preparation, which could have brought up less fertile subsoil. MR and P soils exhibited higher nutrient levels than P + H, attributable to the minimal soil disturbance caused by MR and P in contrast to the P + H soil. Despite a decline in SOM, N, P, K, Ca and Mg over time from their initial values, the decrease in these nutrients was less pronounced in the MC-treated soil compared to MR, P and P + H. Consequently, tillage appears to have accelerated the breakdown of OM, resulting in a greater loss of OM and other nutrients in the tilled plots.

In both locations, the P + H tillage method demonstrated superior outcomes in terms of plant height, number of branches, number of pods and grain yield compared to other

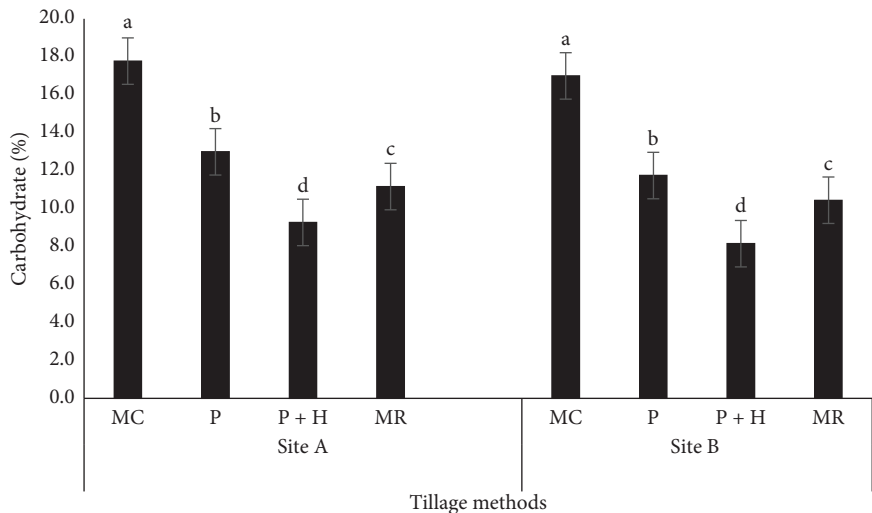


FIGURE 6: Effect of tillage methods on carbohydrate content of soybean. MC = manual clearing; P = ploughing; P + H = ploughing plus harrowing; MR = manual ridging. Vertical bars show standard error of paired comparisons; bars marked with different letters show means significantly different at 5% level using Duncan's multiple range test.

TABLE 6: Effect of tillage methods on mineral contents of soybean.

	Ca (%)		Mg (%)		K (%)		Fe (%)		Na (%)		P (%)		Zn (%)	
	Site A	Site B	Site A	Site B	Site A	Site B	Site A	Site B	Site A	Site B	Site A	Site B	Site A	Site B
MC	0.31d	0.32d	0.29c	0.27d	0.22c	0.20c	0.02b	0.01b	0.03b	0.04b	0.74b	0.75b	0.02b	0.02c
P	0.33cd	0.34cd	0.29c	0.30c	0.23c	0.24b	0.03ab	0.01b	0.04ab	0.05ab	0.80b	0.82b	0.03b	0.04b
P + H	0.44a	0.41a	0.38a	0.40a	0.28a	0.30a	0.04a	0.04a	0.05a	0.06a	0.94a	0.98a	0.04a	0.06a
MR	0.41ab	0.39ab	0.35ab	0.36ab	0.27a	0.25b	0.04a	0.03a	0.05a	0.05b	0.88a	0.89a	0.03ab	0.05ab

Note: Values followed by similar letters under the same column are not significantly different at $p = 0.05$ according to Duncan's multiple range test. Abbreviations: MC = manual clearing; MR = manual ridging; P = ploughing; P + H = ploughing + harrowing.

tillage techniques. This enhanced growth and yield associated with P + H can be attributed to its lower bulk density and higher porosity. Conversely, the higher bulk density resulting from alternative tillage methods may impede root development mechanically, thereby adversely impacting nutrient and water absorption, consequently leading to diminished soybean growth and yield. This observation aligns with the findings of Lasisi and Aluko [12], who noted a significantly better response in soybean growth and seed yield with conventional tillage compared to conservation tillage methods. The explanation provided by Lasisi [15] for this phenomenon points to the higher soil porosity and lower penetration resistance observed in conventionally tilled plots compared to untilled ones. This corroborates earlier studies by Tardieu [46] and Tsegaye and Mullins [47], which associated poor initial root establishment with elevated soil bulk density and penetration resistance.

Despite the higher levels of SOM, N, P, K, Ca and Mg content in the MC treatment (as shown in Table 4), soybean yield was lower than treatments P, MR and P + H (as indicated in Table 5). Furthermore, significant correlations were observed between soil physical properties and yield (Table 7), as well as between soil chemical properties and yield (Table 8), with the exception of soil moisture content, temperature and nitrogen. This suggests that soil moisture content, temperature and nitrogen are not limiting factors

for soybean cultivation in the study area; rather, bulk density appears to exert a more pronounced effect on soybean yield than soil chemical properties. The findings of this study underscore the necessity of employing a suitable degree of tillage for ensuring sustainable soybean production in the derived savanna Alfisol.

The significant elevation in soybean mineral (calcium, magnesium, phosphorus, potassium, iron, sodium, manganese and zinc) and proximate (protein, fat, ash and fibre) levels observed in the P + H tillage method compared to alternative methods could be attributed to the lower bulk density of its soil. This soil condition likely facilitated improved root proliferation, growth and development, fostering robust vegetative growth. Consequently, this enhanced the absorption of nutrients by soybean plants, resulting in an increased number of leaves, heightened photosynthetic activity and improved physiological processes. These factors collectively contribute to a higher production of assimilates, consequently augmenting the chemical composition of soybean leaves and seeds.

Soybean holds significant importance as a food crop widely recognized for its affordability and as a source of vegetable protein within communities [48]. Therefore, any soil management approach that enhances its protein content would be enthusiastically embraced. The increase in protein content in soybeans can be attributed to several factors.

TABLE 7: Correlation coefficient between grain yield, proximate and mineral contents of soybean and soil physical properties.

	Protein	Fat	Ash	Moisture	Fibre	CHO	Ca	Mg	K	Fe	Na	P	Zn	Yield
SMC	-0.96*	-0.97*	-0.96*	0.92	-0.94	0.97*	-0.94	-0.88	-0.95*	-0.99**	-0.99**	-0.96*	-0.87	-0.92
SBD	-0.96*	-0.97*	-0.97*	0.89	-0.96*	0.98*	-0.96*	-0.96*	-0.95*	-0.95*	-0.95*	-0.99**	-0.95*	-0.98*
STP	0.96*	0.97*	0.97*	-0.89	0.95*	-0.98*	0.96*	0.96*	0.95*	0.95*	0.95*	0.99**	0.95*	0.98*
ST	0.86	0.84	0.84	-0.81	0.84	-0.86	0.88	0.81	0.90	0.96*	0.96*	0.85	0.67	0.78

Note: *Significant difference at $p = 0.05$; **significant difference at $p = 0.01$.

Abbreviations: SBD = soil bulk density, SMC = soil moisture content, ST = soil temperature, STP = soil total porosity.

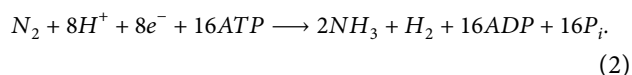
TABLE 8: Correlation coefficient between grain yield, proximate and mineral contents of soybean and soil chemical properties.

	Protein	Fat	Ash	Mois	Fibre	CHO	Ca	Mg	K	Fe	Na	P	Zn	Yield
pH	-0.76	-0.81	-0.80	0.64	-0.68	0.81	-0.88	-0.91	-0.85	-0.74	-0.74	-0.91	-0.91	-0.95*
SOM	-0.91	-0.93	-0.92	0.826	-0.86	0.94	-0.99**	-0.97*	-0.99**	-0.95*	-0.95*	-0.99**	-0.89	-0.97*
N	-0.91	-0.92	-0.91	0.84	-0.87	0.93	-0.97*	-0.92	-0.98*	-0.98*	-0.98*	-0.96*	-0.81	-0.91
P	-0.92	-0.94	-0.93	0.83	-0.87	0.94	-0.99**	-0.96*	-0.98*	-0.96*	-0.96*	-0.99**	-0.90	-0.98*
K	-0.93	-0.96	-0.94	0.85	-0.89	0.93	-0.97*	-0.94	-0.96*	-0.94	-0.94	-0.99**	-0.94	-0.99**
Ca	-0.88	-0.91	-0.90	0.77	-0.81	0.92	-0.97*	-0.97*	-0.96*	-0.89	-0.89	-0.99**	-0.92	-0.99**
Mg	-0.90	-0.91	-0.91	0.80	-0.84	0.93	-0.99**	-0.97*	-0.99**	-0.95*	-0.95*	-0.99*	-0.86	-0.96*

Note: *Significant difference at $p = 0.05$; **significant difference at $p = 0.01$.

Abbreviation: SOM = soil organic matter.

Soybeans rely on sufficient nitrogen to synthesize protein, and optimal nitrogen uptake is facilitated by less compacted soil conditions. One key process in soybeans, vital for meeting nitrogen demands and sustaining protein formation in seeds, is biological nitrogen fixation [49]. Biological nitrogen fixation is a process where atmospheric nitrogen (N_2) is converted into a form that plants can use, such as ammonia (NH_3) [50]. This process is carried out by specific bacteria and archaea, known as diazotrophs, which possess the enzyme nitrogenase. Nitrogenase facilitates the reduction of nitrogen gas to ammonia, a form useable by plants. The overall chemical reaction can be summarized as follows:



This reaction occurs under anaerobic conditions since nitrogenase is highly sensitive to oxygen. Soybeans can also form a mutualistic symbiotic relationship with bacteria from the genus *Rhizobia*. This relationship primarily benefits the soybean by providing it with ammonia derived from atmospheric nitrogen, which the plant can utilize for its growth. In return, *Rhizobia* receive carbohydrates and a protective niche from the plant roots. The symbiosis process involves several steps: (1) Soybean roots release flavonoids into the soil, which attract *Rhizobia*. The bacteria recognize these chemical signals and move towards the root [51]. (2) *Rhizobia* attach to the root hairs and induce the formation of an infection thread, through which they enter the root cells. The plant cells then encapsulate the bacteria in specialized structures called nodules [52, 53]. (3) Inside the nodules, *Rhizobia* differentiate into a form known as bacteroids, which are capable of nitrogen fixation. The plant supplies the bacteroids with carbon compounds derived from photosynthesis, and the bacteroids fix atmospheric nitrogen into ammonia [54]. (4) The nitrogenase enzyme complex within the bacteroids converts atmospheric nitrogen into ammonia.

The ammonia is then assimilated into organic compounds like amino acids and nucleotides, which are transported throughout the plant. (5) The plant provides the *Rhizobia* with carbohydrates and other nutrients, while the *Rhizobia* provide the plant with fixed nitrogen [55]. This nutrient exchange is crucial for the growth and productivity of the soybean plant. This symbiotic relationship not only benefits the soybean but also enriches the soil with nitrogen, enhancing soil fertility and reducing the need for chemical fertilizers. Soil with high bulk density, characterized by low porosity, hampers soil aeration, which is crucial for the activity and survival of nitrogen-fixing bacteria like *Rhizobia*. These bacteria engaged in symbiotic relationships with soybean roots could have their activity impeded in poorly aerated soil due to limited oxygen availability, resulting in reduced nitrogen fixation. Moreover, compacted soil layers pose challenges for soybean roots, hindering their penetration and restricting access to soil moisture, nutrients and symbiotic nitrogen-fixing bacteria. Inadequate root growth and development further diminish the efficiency of establishing symbiotic relationships with nitrogen-fixing bacteria.

The findings regarding the correlation between soil physical and chemical properties and the proximate and mineral contents of soybean are presented in Tables 7 and 8, respectively. In Table 7, significant correlations are observed between soil physical properties, particularly bulk density and porosity, and the proximate content of soybean. However, correlations with soil chemical properties are largely insignificant, especially for the proximate contents. This suggests a greater influence of soil physical properties on the proximate and mineral contents of soybean compared to soil chemical properties. In conditions of MC soil, despite the presence of nutrients, soybean roots face challenges in nutrient absorption during early growth stages due to the high density and low porosity of the soil. This impedes root proliferation, consequently affecting the growth, yield and quality of soybean. Gole [56] noted a significant increase in

soybean yield with subsurface tillage, attributing it to improved moisture and nutrient availability facilitated by a more extensive root system developed postsubsoiling, leading to enhanced vegetative and reproductive growth. Studies by Whiteley and Dexter [57] demonstrated that untilled soil significantly slows down the elongation rate of soybean seminal roots by about 50% compared to tilled soil. Additionally, nontillage practices were found to reduce seedling emergence in wheat (*Triticum aestivum*) and pea crops by approximately 10% across seven instances compared to tilled soil. In fact, root penetration into soil is inhibited when bulk density exceeds 1.5–1.8 g/cm³ according to Russell [58] and Maurya and Lal [59]. The average bulk density of MC soil in this experiment was measured at 1.58 g/cm³. That means root growth and nutrient absorption in MC soil will have been greatly inhibited.

Studies have shown that the most important factor leading to an increase in plant yields is the rational application of fertilizers [60–63].

5. Conclusion

Tillage methods influenced soil physical and chemical properties, yield, growth, proximate and mineral contents of soybean significantly. This research demonstrates that in the Alfisols of a derived savanna, tillage is essential for successful soybean growth. Specifically, P + H effectively decreases soil bulk density, promoting the proliferation, growth and development of both primary and secondary roots. This, in turn, fosters robust vegetative growth and enhances nutrient absorption by soybean plants, resulting in higher seed yield and improved proximate and mineral contents. The increasing order of growth and yield of soybean was as follows: MC < P < MR < P + H. Using the mean of the two sites, relative to MC, P and MR, P + H increased the grain yield of soybean by 41.3, 24.4 and 13.7%, respectively. On the other hand, MC leads to higher soil bulk density than tilled soils, particularly P + H, resulting in significant yield loss and decreased soybean quality. The increasing order of soil bulk density among the various tillage methods was as follows: P + H < MR < P < MC. The reverse order was the porosity, while that of soil chemical properties exhibited by the different tillage systems was as follows: P + H < MR < P < MC. Despite MC exhibiting higher soil nutrient concentrations and OM status, these advantages do not translate into increased yield and quality due to its higher bulk density. For small-scale operations, MR is suggested, while P + H is recommended for large-scale soybean production. Future studies could include additional variables, such as economic analysis, long-term soil health impacts or the effects of tillage methods under varying climate conditions. This would provide a more comprehensive understanding of the sustainability of the recommended practices.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Funding

No funding was received for this manuscript.

References

- [1] A. O. Adekiya, T. M. Agbede, and A. O. Ojomo, "Effect of Tillage Methods on Soil Properties, Nutrient Contents, Growth and Yield of Tomato on an Alfisol of Southwestern Nigeria," *American-Eurasian Journal of Sustainable Agriculture* 3, no. 3 (2009): 348–353.
- [2] A. O. Adekiya and S. O. Ojeniyi, "Evaluation of Tomato Growth and Soil Properties Under Methods of Seedling Bed Preparation in an Alfisol in the Rainforest Zone of Southwest Nigeria," *Soil and Tillage Research* 64, no. 3–4 (2002): 275–279, [https://doi.org/10.1016/s0167-1987\(01\)00232-x](https://doi.org/10.1016/s0167-1987(01)00232-x).
- [3] A. O. Adekiya, S. O. Ojeniyi, and T. M. Agbede, "Soil Physical and Chemical Properties and Cocoyam Yield Under Different Tillage Systems in a Tropical Alfisol," *Experimental Agriculture* 47, no. 3 (2011): 477–488, <https://doi.org/10.1017/s001447971100041x>.
- [4] O. Adugna, "Effect of Different Tillage Practices on Production of Soya Bean-Maize (*Zea mays* L. *Glycine max* L.) in Clay Loam of Assosa, Ethiopia," *International Journal of Environmental Sciences and Natural Resources* 19, no. 5 (2019): 556023, <https://doi.org/10.19080/IJESNR.2019.19.556023>.
- [5] B. A. Senjobi, O. T. Ande, and A. E. Okulaja, "Effects of Tillage Practices on Soil Properties Under Maize Cultivation on Oxic Paleustalf in South Western Nigeria," *Open Journal of Soil Science* 03, no. 03 (2013): 163–168, <https://doi.org/10.4236/ojss.2013.33019>.
- [6] M. A. Busari, S. S. Kukal, A. Kaur, R. Bhatt, and A. A. Dulazi, "Conservation Tillage Impacts on Soil, Crop and the Environment," *International Soil and Water Conservation Research* 3, no. 2 (2015): 119–129, <https://doi.org/10.1016/j.iswcr.2015.05.002>.
- [7] M. S. Kahlon, R. Lal, and M. Ann-Varughese, "Twenty-Two Years of Tillage and Mulching Impacts on Soil Physical Characteristics and Carbon Sequestration in Central Ohio," *Soil and Tillage Research* 126, no. 2 (2013): 151–158, <https://doi.org/10.1016/j.still.2012.08.001>.
- [8] T. M. Agbede and A. O. Adekiya, "The Effect of Tillage on Soil Physical and Chemical Properties and Yield of Ginger," *Agriculturae Conspectus Scientifici* 83, no. 4 (2018): 315–320.
- [9] T. M. Agbede and A. O. Adekiya, "Soil Properties and Yam Yield Under Different Tillage Systems in A Tropical Alfisol," *Archives of Agronomy and Soil Science* 59, no. 4 (2013): 505–519, <https://doi.org/10.1080/03650340.2011.652096>.
- [10] S. H. M. Aikins and J. J. Afuakwa, "Effect of Four Different Tillage Practices on Cowpea Performance," *World Journal of Agricultural Sciences* 6, no. 6 (2010): 644–651.
- [11] T. M. Agbede, "Effect of Tillage on Soil Properties and Yam Yield on an Alfisol in Southwestern Nigeria," *Soil and Tillage Research* 86 (2006): 1–8, <https://doi.org/10.1016/j.still.2005.01.012>.
- [12] D. Lasisi and O. B. Aluko, "Effects of Tillage Methods on Soybean Growth and Yield in a Tropical Sandy Loam Soil," *International Agrophysics* 23 (2009): 147–153.
- [13] A. B. Chorey, S. J. Dhage, N. K. Darekar, and V. D. Hedao, "Effect of Various Tillage Practices on Soybean Productivity

- and Soil Moisture Dynamics Under Rainfed Condition," *International Journal of Current Microbiology and Applied Sciences* 11 (2020): 3231–3236.
- [14] E. A. Ajav and A. O. Oyelami, "The Effect of Mechanical Induced Compaction on Soybean Growth for Two Soils of Nigeria," *Proceedings of the International Conference on Industrial* (2001).
 - [15] D. Lasisi, *Effects of Tillage Methods on Soybean Growth and Soil Properties* (Ile-Ife, Nigeria: Obafemi Awolowo University, 2008).
 - [16] F. O. Adekayode, "Appropriate Tillage Methods to Minimize Soil Degradation and Improve the Yield of Soybean (*Glycine max.*) in Southwestern Nigeria," *Scientific Research Annals* 1, no. 1 (2004): 41–45.
 - [17] A. O. Adekiya, C. M. Aboyeji, O. Dunsin, et al., "Poultry Manure Addition Affects Production, Plant Nutritional Status and Heavy Metals Accumulation in Green Amaranth (*Amaranthus hybridus*)," *International Journal of Agriculture and Biology* 22, no. 5 (2019): 993–1000.
 - [18] D. J. Campbell and J. K. Henshall, "Bulk Density," in *Physical Methods of Soil Analysis*, eds. K. A. Smith and C. E. Mullins (New York: Marcel Dekker, 1991), 329–366.
 - [19] G. W. Gee and D. Or, "Particle-Size Analysis," in *Methods of Soil Analysis. Part 4. Physical Methods*, eds. J. H. Dane and G. C. Topp (Madison: Soil Science Society of America, 2002), 255–293.
 - [20] N. Hunt and R. Gilkes, *Farm Monitoring Handbook* (Netherlands: The University of Western Australia, 1992).
 - [21] N. C. Brady and R. R. Weil, *The Nature and Properties of Soils*, 12th edition (New Jersey: Prentice-Hall, 1999).
 - [22] M. R. Carter and E. G. Gregorich, "Soil Sampling and Methods of Analysis," *Canadian Society of Soil Science*, 2nd edition (Boca Raton: CRC Press and Taylor and Francis Group, 2007).
 - [23] D. W. Nelson and L. E. Sommers, "Total Carbon, Organic Carbon and Organic Matter," in *Methods of Soil Analysis. Part 3*, 2nd edition, ed. D. L. Sparks (ASA and SSSA, 1996), 961–1010.
 - [24] J. M. Bremner, "Nitrogen-Total," in *Methods of Soil Analysis. Part 3. Chemical Methods*, 2nd edition, ed. D. I. Sparks (Madison: ASA and SSSA, 1996).
 - [25] L. P. Van Reeuwijk, *Procedures for Soil Analysis*, 6th edition (Wageningen, Netherlands: International Soil Reference and Information Centre (ISRIC) and Food and Agriculture Organization of the United Nations (FAO), 2002).
 - [26] W. H. Hendershot, H. Lalonde, and M. Duquette, "Ion Exchange and Exchangeable Cations," in *Soil Sampling and Methods of Analysis*, 2nd edition, eds. M. R. Carter and E. G. Gregorich (Boca Raton: CRC Press and Taylor and Francis Group, 2008), 173–178.
 - [27] A. A. Ibitoye, *Laboratory Manual on Basic Soil Analysis*, 2nd edition (Akure: Foladaye Publishing Company, 2006).
 - [28] AOAC, "Official Methods of Analysis," in *Association of Official Agricultural Chemists International* (2010).
 - [29] E. J. Udo and J. A. Oguwele, "Laboratory Manual for the Analysis of Soil, Plants, and Water Samples," *Department of Crop Production*, 3rd edition (Kwara State, Nigeria: University of Ilorin, 1986).
 - [30] A. F. Gabriel, N. C. Igwemmar, A. A. Sadam, and S. A. Babalola, "Characterization of Seed Oil From *Citrullus lanatus* (Watermelon)," *Direct Res. J. Public Health and Environ. Technol.* 3, no. 2 (2018): 34–40.
 - [31] L. C. De Conto, M. A. L. Gragnani, D. Maus, et al., "Characterization of Crude Watermelon Seed oil by Two Different Extractions Methods," *Journal of the American Oil Chemists' Society* 88, no. 11 (2011): 1709–1714, <https://doi.org/10.1007/s11746-011-1850-8>.
 - [32] I. Shahid, A. V. Chauhdry, and M. A. Anjum, "Effect of Root Stock on Leaf Mineral Composition and Productivity of Kinnow Mandarin," *International Journal of Agriculture and Biology* 1 (1999): 91–93.
 - [33] Genstat, *Genstat 5 Release 3.2 Reference Manual* (Oxford, UK: Oxford University Press, 1993).
 - [34] E. A. Akinrinde and G. O. Obigbesan, "Evaluation of the Fertility Status of Selected Soils for Crop Production in Five Ecological Zones of Nigeria," in *Proceedings of the 26th Annual Conference of Soil Science Society of Nigeria*, ed. O. Babalola (2000), 279–288.
 - [35] E. A. Bankole and S. O. Ojeniyi, "Comparative Effect of Ridge Furrow and Zero Tillage on Cowpea at Abeokuta and Akure, Nigeria," *Nigerian Journal of Soil Science* 15 (2005): 54–59.
 - [36] I. F. Ike, "Soil and Crop Responses to Different Tillage Practices in a Ferruginous Soil in the Nigerian Savanna," *Soil and Tillage Research* 6, no. 3 (1986): 261–272, [https://doi.org/10.1016/0167-1987\(86\)90460-5](https://doi.org/10.1016/0167-1987(86)90460-5).
 - [37] R. L. Donahue, R. W. Miller, and J. C. Shickluna, "Soils," *An Introduction to Soils and Plant Growth*, 5th edition (1990).
 - [38] S. O. Ojeniyi, B. Abiwonnu, and S. A. Odedina, "Evaluation of Soil and Maize Performance in Ridge and Untilled Seedbeds at Akure, Southwest Nigeria," *Nigeria Journal of Soil Science* 16 (2006): 131–135.
 - [39] S. O. Ojeniyi, "Soil Structure Produced by Tillage Implements and Their Effects on Soil Water," *Ghana Journal of Agricultural Science* 14 (1981): 45–49.
 - [40] R. Lal, "Effects of 6 Years of Continuous No-Till or Puddling Systems on Soil Properties and Rice (*Oryza sativa*) Yield of a Loamy Soil," *Soil and Tillage Research* 8 (1986): 181–200, [https://doi.org/10.1016/0167-1987\(86\)90333-8](https://doi.org/10.1016/0167-1987(86)90333-8).
 - [41] S. O. Ojeniyi, A. O. Adetoro, and S. A. Odedina, "Effect of Manual Tillage Methods on Soil Density, Moisture Content and Cowpea Grain Yield," *Journal of Agricultural Technology* 7 (1999): 16–19.
 - [42] N. U. Ndaeyo, E. A. Aiyelari, and A. A. Agboola, "Evaluation of Different Tillage Practices for Monocultural Cowpea (*Vigna unguiculata* (L) Walp) Production in Ibadan Southwestern Nigeria," *African soil* 28 (1995): 475–488.
 - [43] T. M. Agbede, "Nutrient Availability and Cocoyam Yield Under Different Tillage Practices," *Soil and Tillage Research* 99, no. 1 (2008): 49–57, <https://doi.org/10.1016/j.still.2007.12.003>.
 - [44] T. M. Agbede and S. O. Ojeniyi, "Tillage and Poultry Manure Effects on Soil Fertility and Sorghum Yield in Southwestern Nigeria," *Soil and Tillage Research* 104, no. 1 (2009): 74–81, <https://doi.org/10.1016/j.still.2008.12.014>.
 - [45] A. O. Adekiya, T. M. Agbede, S. O. Ojeniyi, and B. S. Ewulo, "Influence of Tillage and Poultry Manure on Soil Properties and Cormel Yield of Cocoyam," *Proceedings of the International Soil Tillage Research Organisation (ISTRO) Nigeria Symposium* (Akure, Nigeria: Akure, 2014).
 - [46] F. Tardieu, "Growth and Functioning of Roots and of Root Systems Subjected to Soil Compaction. Towards a System With Multiple Signalling?" *Soil and Tillage Research* 30, no. 2–4 (1994): 217–243, [https://doi.org/10.1016/0167-1987\(94\)90006-x](https://doi.org/10.1016/0167-1987(94)90006-x).
 - [47] T. Tsegaye and C. E. Mullins, "Effect of Mechanical Impedance on Root Growth and Morphology of Two Varieties of Pea (*Pisum sativum* L.)," *New Phytologist* 126, no. 4 (1994): 707–713, <https://doi.org/10.1111/j.1469-8137.1994.tb02965.x>.
 - [48] A. A. Kurniawan, P. B. Hastuti, and A. Umami, "Growth and Yield of Soybeans in Various Growing Media Composition

- and Inoculation of Rhizobacteria on Marginal Soils,” *Agro-technology Research Journal* 5, no. 2 (December 2021): 104–109, <https://doi.org/10.20961/agrotechresj.v5i2.53968>.
- [49] I. A. Ciampitti, A. F. de Borja Reis, S. C. Córdova, et al., “Revisiting Biological Nitrogen Fixation Dynamics in Soybeans,” *Frontiers in Plant Science* 12 (2021): 727021, <https://doi.org/10.3389/fpls.2021.727021>.
- [50] S. C. Wagner, “Biological Nitrogen Fixation,” *Nature Education Knowledge* 3, no. 10 (2011): 15.
- [51] A. Shah and D. L. Smith, “Flavonoids in Agriculture: Chemistry and Roles in, Biotic and Abiotic Stress Responses, and Microbial Associations,” *Agronomy* 10, no. 8 (2020): 1209, <https://doi.org/10.3390/agronomy10081209>.
- [52] A. V. Tsyganova, N. J. Brewin, and V. E. Tsyganov, “Structure and Development of the Legume-Rhizobial Symbiotic Interface in Infection Threads,” *Cells* 10, no. 5 (May 2021): 1050, <https://doi.org/10.3390/cells10051050>.
- [53] M. Libault, L. Brechenmacher, J. Wan, and G. D. May, “Genetics and Functional Genomics of Legume Nodulation Current,” *Opinion in Plant Biology* 9, no. 2 (April 2006): 110–121.
- [54] E. J. Patriarca, R. Tatè, and M. Iaccarino, “Key Role of Bacterial NH_4^+ Metabolism in Rhizobium-Plant Symbiosis,” *Microbiology and Molecular Biology Reviews* 66, no. 2 (Jun 2002): 203–222, <https://doi.org/10.1128/mmbr.66.2.203-222.2002>.
- [55] W. Lindemann and C. Glover, *Nitrogen Fixation by Legumes* (New Mexico: New Mexico State University, 2015), <https://aces.nmsu.edu>.
- [56] S. M. Gole, “Effect of Different Tillage Practices on Growth and Yield of Cotton Based Cropping System” (2012).
- [57] G. M. Whiteley and A. R. Dexter, “Root Development and Growth of Oilseed, Wheat and Pea Crops on Tilled and Non-Tilled Soil,” *Soil and Tillage Research* 2, no. 4 (1982): 379–393, [https://doi.org/10.1016/0167-1987\(82\)90006-x](https://doi.org/10.1016/0167-1987(82)90006-x).
- [58] E. W. Russel, *Soil Conditions and Plant Growth*, 10th edition (London, UK: Longman, 1973).
- [59] P. R. Maurya and R. Lal, “Effects of Bulk Density and Soil Moisture on Radicle Elongation of Some Tropical Crops,” in *Soil Physical Properties and Crop Production in the Tropics*, eds. R. Lal and D. J. Greenland (New York, NY: Wiley, 1979), 339–347.
- [60] V. Popović, S. Vučković, Z. Jovović, et al., “Genotype by Year Interaction Effects on Soybean Morpho-Productive Traits and Biogas Production,” *Genetika* 52, no. 3 (2020): 1055–1073, <https://doi.org/10.2298/GENSR2003055P>.
- [61] L. Kolarić, V. Popović, L. Živanović, et al., “Buckwheat Yield Traits Response as Influenced by Row Spacing, Nitrogen, Phosphorus, and Potassium Management,” *Agronomy* 11 (2021): 2371, <https://doi.org/10.3390/agronomy11122371>.
- [62] V. Vasileva, G. Georgiev, and V. Popović, “Genotypic Specificity of Soybean [Glycine Max (L.) Merr.] Plastid Pigments Content Under Sowing Date and Interrow Spacing,” *Genetika* 55, no. 2 (2023): 455–471, <https://doi.org/10.2298/GENSR2302455V>.
- [63] V. Miladinović, V. Popović, D. Delić, et al., “Effectiveness of Bradyrhizobium Japonicum Inoculation on Nodulation Dynamics in Glycine Max (L.) Roots,” *Notulae Botanicae Horti Agrobotanici Cluj-Napoca* 52, no. 2 (2024): 13627, <https://doi.org/10.15835/nbha52213627>.